ANSWER SCHEME
PSPM 2019/2020

No.		Answer
1.	(a)	i.

	$^{79}_{35}Br$	$^{81}_{35}Br$
Proton	35	35
Electron	35	35
Neutron	44	46

Isotopes are atoms of the same element with the same protons number but different number of neutrons.

ii.

Br^-

No of electrons = 36

(b) Mass of substance = 1.0 g

CO_2 = 2.52g

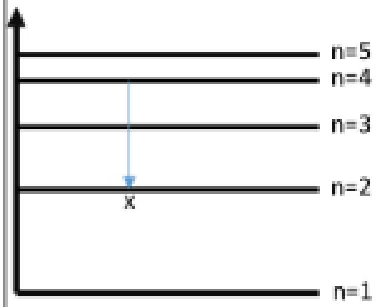
H_2O = 0.443g

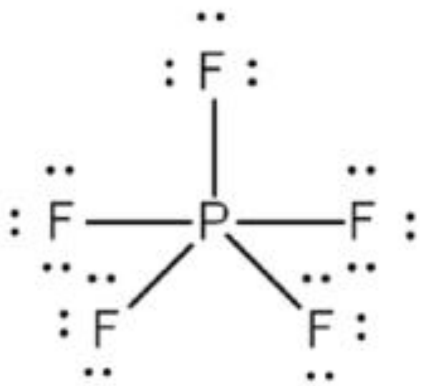
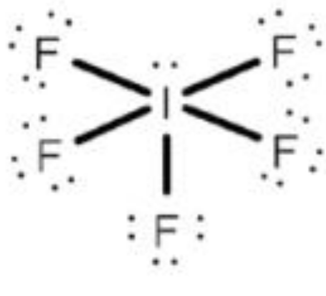
$n_{CO_2} = \frac{2.52g}{44\ gmol^{-1}}$ $= 0.0573\ mol$	$n_{H_2O} = \frac{0.443g}{18\ gmol^{-1}}$ $= 0.0246\ mol$
$1\ mol\ CO_2 \equiv 1\ mol\ C$ $0.0573\ mol\ CO_2 \equiv 0.0573\ mol\ C$ $Mass\ C = 0.0573\ mol\ x\ 12\ gmol^{-1}$ $= 0.06872\ g$	$1\ mol\ H_2O \equiv 2\ mol\ H$ $0.0246\ mol\ H_2O \equiv 0.0492\ mol\ H$ $Mass\ H = 0.0492\ mol\ x\ 1\ gmol^{-1}$ $= 0.0492\ g$

$$mass\ O = 1.0g - (0.6872g + 0.0492g)$$
$$= 0.2636g$$

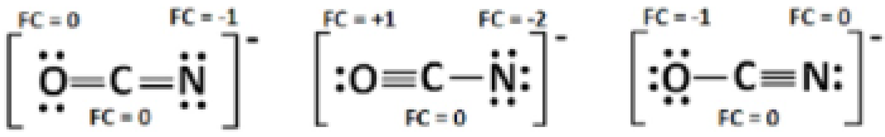
Element	C	H	O
Mass (g)	$= 0.6872g$	$= 0.0492g$	$= 0.2636g$
Mole (mol)	$= \frac{0.6872g}{12\ gmol^{-1}}$ $= 0.0573\ mol$	$= \frac{0.0492g}{1\ gmol^{-1}}$ $= 0.0492\ mol$	$= \frac{0.2636g}{16\ gmol^{-1}}$ $= 0.0165\ mol$
Simplest ratio	$= \frac{0.0573\ mol}{0.0165\ mol}$ $= [3.47]x2$ $= 7$	$= \frac{0.0492\ mol}{0.0165\ mol}$ $= [2.98]x2$ $= 6$	$= \frac{0.0165\ mol}{0.0165\ mol}$ $= [1]x2$ $= 2$
Emprical formula	$C_7H_6O_2$		

No.		Answer
	(c)	$Mr_{Ca(C_2H_3O_2)_2} = 158.1 gmol^{-1}$ $V_{solution} = 250\ mL$ $\rho = 1.509 gmL^{-1}$ $molality = ?$ $\rho_{solution} = \frac{mass_{solution}}{V_{solution}}$ $Mass_{solution} = 1.509\ gmL^{-1} x 250\ mL$ $Mass_{solution} = 377.25\ g$ $Molarity = \frac{n_{solute}}{V_{solution(L)}}$ $n_{solute} = 0.25\ M x\ 0.250\ L$ $= 0.0625\ mol$ $Mass_{solute} = 0.0625\ M x\ 158\ gmol^{-1}$ $= 9.88\ g$ $Mass_{solvent} = 377.25\ g - 9.88\ g$ $= 367.25\ g$ $Molality = \frac{n_{solute}}{Mass_{sovent(kg)}}$ $Molality = \frac{0.0625\ mol}{0.367\ kg}$ $= 0.1703\ molal$
	(d)	$MgCl_2 + 2NaOH \rightarrow 2NaCl + Mg(OH)_2$ $Mass_{MgCl_2} = 15.1\ g$ $Mass_{NaOH} = 9.35\ g$ $n_{MgCl_2} = \frac{15.1\ g}{95.3\ gmol^{-1}}$ $= 0.158\ mol$ $n_{NaOH} = \frac{9.53\ g}{40.0\ gmol^{-1}}$ $= 0.2338\ mol$ $(n_{NaOH}\ available)$ $1\ mol\ MgCl_2 \equiv 2\ mol\ NaOH$ $0.158\ mol\ MgCl_2 \equiv 0.316\ mol\ NaOH$ $(n_{NaOH}\ needed)$ $n_{NaOH}\ available\ < n_{NaOH}\ needed$ $\therefore NaOH\ ia\ a\ limiting\ reactant$

No.		Answer
		$2\ mol\ NaOH \equiv 1\ mol\ Mg(OH)_2$ $0.2338\ mol\ NaOH \equiv \frac{1}{2}\ mol\ Mg(OH)_2$ $= 0.1169\ mol\ Mg(OH)_2$ $Mass_{Mg(OH)_2} = 0.1169\ mol\ x\ 58.3\ \ gmol^{-1}$ $= 6.815\ g$
2.	(a)	i. Balmer series
		ii.  Line X produced when an electron at ground state gain energy. It will excite to higher energy level (n=4). At higher energy level, electron is unstable. It will fall back to lower energy level (n=2) and produce a second line in Balmer series and emitted a light in a formed of photons.
		iii. $\frac{1}{\lambda} = R_H\left(\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right), n_1 < n_2$ $\frac{1}{\lambda} = 1.097x10^7\left(\frac{1}{2^2} - \frac{1}{4^2}\right)$ $\lambda = 486\ nm$
		iv. $\Delta E = \frac{hc}{\lambda}$ $= \frac{(6.6256x10^{-34})(3.0x10^8)}{486x10^{-9}}$ $= 4.098x10^{-19}J$

No.		Answer
	(b)	Anomalous in $_{24}Cr$ $_{24}Cr$ expected configuration  $_{24}Cr$ actual configuration  **3d** electronic configuration in expected configuration is partially filled while the actual is half-filled. Half-filled electronic configuration is more stable than partially filled.
3.	(a)	i. • Both O and S are elements from group 16. • O is an element from period 2. It cannot expend the octet. $F - \ddot{\underset{\cdot\cdot}{O}} - F$ • While S is an element from period 3. Thus it can expand their octet. • O can only use p orbital • S has an empty d orbital
		ii.   Both bonded to 5 fluorine. PF_5 have equal repulsion according to VSEPR theory. Thus, the shape is trigonal bipyramidal. 5 bonding pair. IF_5 has 5 bonding pair + 1 lone pair. Thus the shape become square pyramidal.

No.		Answer
		ii.

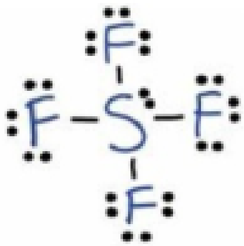


I II III

Structure III is more plausible because it has low formal charge. The negative charge (-1) is on more electronegative atom (oxygen).

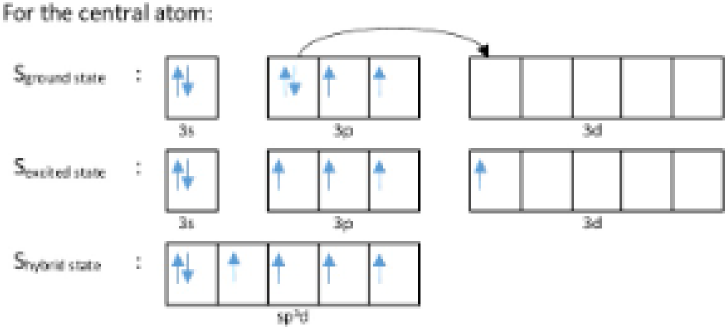
(b)

S	=1x6	=6
F	=4x7	=28
Minus electron bond		=34 – 8
Minus lone pair electron		=26 – 24
Extra electron		=2

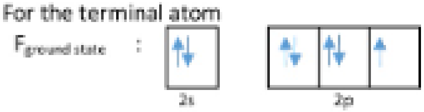


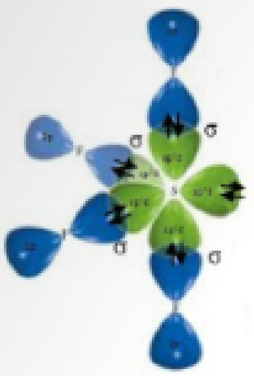
1 lone pair + 4 bonding pair
4 sigma + 1 lone pair
sp^3d

For the central atom:



For the terminal atom



No.		Answer
		
	(c)	• Group 1 have metallic bond between atoms / particles. • Group 17 have Van der Waals between molecules. • Metallic bond is stronger than Van der Waals forces. • More energy required to overcome the metallic bonds. • Thus melting point Group 1> Group 17 •
4.	(a)	$S(s) + O_2(g) \rightarrow SO_2(g)$ $n_{sulphur} = \frac{2.54x10^3\ g}{32.1\ gmol^{-1}}$ $= 79.128\ mol$ T= 30.5 + 273.15 = 303.65K P=851.2mmHg = 1.12 atm $1\ mol\ S \equiv 1\ mol\ SO_2$ $79.128\ mol\ S \equiv 79.128\ mol\ SO_2$ $PV = nRT$ $V = \frac{nRT}{P}$ $V = \frac{(79.128)(0.08206)(303.65)}{1.12}$ $V = 1760.43\ L$
	(b)	T= 24.0 + 273.15 = 297.15 K P_T=851.2 mmHg V=128 mL = 0.128 L $P_{NH_2} = 762\ mmHg - 22.4\ mmHg$ $P_{NH_2} = 739.6\ mmHg$ $P_{NH_2} = 0.973\ atm$

No.		Answer

$$m = \frac{PVMr}{RT}$$
$$m = \frac{(0.973)(0.128)(17.0)}{(0.08206)(297.15)}$$
$$m = 0.0868\ g$$

5. (a)

	$2SO_2(g)$	$\rightleftharpoons$	$SO_2(g)$	+	$O_2\ (g)$
n_i/mol	1.0		0		0
Δn/mol	-2x		+2x		+x
n $\rightleftharpoons$ /mol	1.0-2x =0.6 mol		2x =0.4 mol		x =0.2 mol
[ ] $\rightleftharpoons$ /M	$= \frac{0.6\ mol}{2\ L}$ $= 0.3$ M		$= \frac{0.4\ mol}{2\ L}$ $= 0.2$ M		$= \frac{0.2\ mol}{2\ L}$ $= 0.1\ M$

$$1 - x = 0.6\ mol$$
$$x = 0.2\ mol$$

$$K_C = \frac{[SO_2]^2[O_2]}{[SO_3]^2}$$
$$K_C = \frac{(0.2)^2(0.1)}{(0.3)^2}$$
$$K_C = 0.044$$

(b) i.

When the temperature decrease, the system will shift to the right. Since it is an exothermic process. Thus, K_p increase.

ii.

Adding inert gas does not affect the equilibrium position. Since the partial pressure of each gas does not change when argon added at constant volume.

6. (a) i.

	$HNO_2(aq)$	+	$H_2O(l)$	$\rightleftharpoons$	$H_3O^+(aq)$	+	$NO_2^-(g)$
[ ]$_i$/M	0.215M				0		0
Δ[ ]/M	-x				+x		+x
[ ] $\rightleftharpoons$ /M	0.215-x				x		x
[ ] $\rightleftharpoons$ /M	$= 0.215M - 8.6x10^{-3}M$ $= 0.2064\ M$				$= 8.6x10^{-3}M$		$= 8.6x10^{-3}M$

$$\%dissociation = \frac{[\ \ ]_{dissociate}}{[\ \ ]_{initial}} x100$$
$$4\% = \frac{x}{0.215M} x100$$
$$x = 8.6x10^{-3}M$$

<table>
<tr><th colspan="2">No.</th><th>Answer</th></tr>
<tr><td></td><td></td><td>ii.
$$K_a = \frac{[H_3O^+]^2[NO_2^-]}{[HNO_2]}$$
$$K_a = \frac{(8.6x10^{-3})^2}{0.2064}$$
$$K_a = 3.58\ x10^{-4}$$</td></tr>
<tr><td></td><td></td><td>iii.
$$pH = -\log[H_3O^+]$$
$$pH = -\log\left(8.6x10^{-3}\right)$$
$$pH = 2.065$$</td></tr>
<tr><td></td><td>(b)</td><td><table>
<tr><td></td><td>$Ca(OH)_2(aq)$ $\rightleftharpoons$</td><td>$Ca^{2+}(aq)$ +</td><td>$2OH^-$ (aq)</td></tr>
<tr><td>[ ]$_i$/M</td><td>0.3</td><td>0</td><td>0</td></tr>
<tr><td>[ ] $\rightleftharpoons$ /M</td><td>0</td><td>0.3</td><td>0.3 x 2
=0.6M</td></tr>
</table>
$$pOH = -\log[OH^-]$$
$$pOH = -\log(0.6)$$
$$pOH = 0.22$$
$$pH = 14 - 0.22$$
$$pH = 13.78$$</td></tr>
<tr><td></td><td>(c)</td><td>i.
$HCN(aq) + KOH(aq) \rightarrow KCN(aq) + H_2O(l)$</td></tr>
<tr><td></td><td></td><td>ii.
M_{KOH}= 0 M</td></tr>
<tr><td></td><td></td><td>iii.
$KCN(s) \rightarrow K^+(aq) + CN^-(aq)$

$K^+(aq) + H_2O(l) \rightarrow$ cannot hydrolysed

$CN^-(aq) + H_2O(l) \rightleftharpoons HCN(aq) + OH^-(aq)$
CN^- is a strong conjugate base. The hydrolysis CN^- with water produce OH^-. Therefore KCN is a basic salt.</td></tr>
<tr><td></td><td></td><td>iii.

Phenolphtalein
Tymol blue</td></tr>
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